ECEN 222 Midterm 1 Cheat Sheet

Lectures 1-7: Signals, Passives, Time/Freq Domain, Diodes

Lecture 1: Signals

~~Thevenin & Norton Sources~~

Equivalence: $v_s(t) = R_s \cdot i_s(t)$

Thevenin voltage divider: $v_o = v_s \frac{R_L}{R_L+R_s}$

Norton current divider: $i_o = i_s \frac{R_s}{R_s+R_L}$

Good voltage source: $R_s \ll R_L$

Good current source: $R_s \gg R_L$

Sinusoidal Signals

$v(t) = V_a \sin(\omega t)$ or $V_a \cos(\omega t)$

Parameters:

- V_a = amplitude (peak)
- $\omega = 2\pi f$ = angular frequency (rad/s)
- $f = 1/T$ = frequency (Hz)
- T = period (s)
- $V_{rms} = V_a/\sqrt{2}$

Fourier Series

Any periodic signal can be represented as sum of sinusoids.

Square wave: $v(t) = \frac{4V}{\pi}\left(\sin\omega_0 t + \frac{1}{3}\sin 3\omega_0 t + \frac{1}{5}\sin 5\omega_0 t + \cdots\right)$

where $\omega_0 = 2\pi/T$

Lecture 2: Passives

Resistor

$V = IR$ (Ohm's Law)

$P = I^2R = V^2/R$ (power dissipated)

$R = \frac{l}{\sigma A}$ (resistance from geometry)

Capacitor

I-V Relation: $i = C\frac{dv}{dt}$

Integrated form: $v(t) = \frac{1}{C}\int_{t_0}^{t} i(\tau)d\tau + v(t_0)$

Stored energy: $W = \frac{1}{2}CV^2 = \frac{1}{2}QV$

Capacitance: $C = \frac{\epsilon_0 A}{d}$ (parallel plate)

Charge: $Q = CV$

Key property: Voltage cannot change instantaneously

Inductor

I-V Relation: $v = L\frac{di}{dt}$

Integrated form: $i(t) = \frac{1}{L}\int_{t_0}^{t} v(\tau)d\tau + i(t_0)$

Stored energy: $W = \frac{1}{2}LI^2$

Inductance: $L = \frac{\mu_0 N^2 A}{l}$ (solenoid)

Key property: Current cannot change instantaneously

Lecture 3: Time Domain

First-Order Circuits

General solution: $x(t) = x(\infty) + [x(0^+) - x(\infty)]e^{-t/\tau}$

where x is v_C or i_L

RC Circuits

Time constant: $\tau = RC$

Source-free (discharge): $v_C(t) = V_0 e^{-t/RC}$

$i(t) = \frac{V_0}{R}e^{-t/RC}$

Step response (charging): $v_C(t) = V_s(1 - e^{-t/RC})$

$i(t) = \frac{V_s}{R}e^{-t/RC}$

RL Circuits

Time constant: $\tau = L/R$

Source-free (discharge): $i_L(t) = I_0 e^{-Rt/L}$

$v_L(t) = -RI_0 e^{-Rt/L}$

Step response (charging): $i_L(t) = \frac{V_s}{R}(1 - e^{-Rt/L})$

$v_L(t) = V_s e^{-Rt/L}$

Key Time Constants

At $t = \tau$: reaches 63.2% of final value

At $t = 5\tau$: essentially at steady state (> 99%)

Second-Order RLC Circuits

Natural frequency: $\omega_0 = \frac{1}{\sqrt{LC}}$

Damping coefficient: $\alpha = \frac{R}{2L}$

Differential equation: $\frac{d^2 v_C}{dt^2} + \frac{R}{L}\frac{dv_C}{dt} + \frac{1}{LC}v_C = 0$

Lecture 4: Complex Numbers

Forms

Rectangular: $z = a + jb$

Polar: $z = r\angle\theta = re^{j\theta}$

Conversions

Rect → Polar:

$r = |z| = \sqrt{a^2 + b^2}$

$\theta = \tan^{-1}(b/a)$

Polar → Rect:

$a = r\cos(\theta)$

$b = r\sin(\theta)$

Euler's Formula

$e^{j\theta} = \cos(\theta) + j\sin(\theta)$

$\cos(\theta) = \frac{e^{j\theta}+e^{-j\theta}}{2}$

$\sin(\theta) = \frac{e^{j\theta}-e^{-j\theta}}{2j}$

Operations

Addition (use rectangular): $(a_1 + jb_1) + (a_2 + jb_2) = (a_1 + a_2) + j(b_1 + b_2)$

Multiplication (use polar): $r_1 e^{j\theta_1} \cdot r_2 e^{j\theta_2} = (r_1 r_2)e^{j(\theta_1+\theta_2)}$

Division (use polar): $\frac{r_1 e^{j\theta_1}}{r_2 e^{j\theta_2}} = \frac{r_1}{r_2}e^{j(\theta_1-\theta_2)}$

Complex Conjugate

$z = a + jb \Rightarrow z^* = a - jb$

$z = re^{j\theta} \Rightarrow z^* = re^{-j\theta}$

$z \cdot z^* = |z|^2 = r^2$

Lecture 5: Frequency Domain

Phasor Transform

$v(t) = V_m\cos(\omega t + \phi) \Leftrightarrow \mathbf{V} = V_m\angle\phi$

Key insight: $\frac{d}{dt} \to j\omega$ (multiplication)

$\int dt \to \frac{1}{j\omega}$ (division)

Impedance

$\mathbf{Z} = \frac{\mathbf{V}}{\mathbf{I}} = R + jX$

$|\mathbf{Z}| = \sqrt{R^2 + X^2}$

$\theta = \tan^{-1}(X/R)$

Element Impedances

Resistor: $\mathbf{Z}_R = R$ (phase: 0°)

Inductor: $\mathbf{Z}_L = j\omega L$ (phase: +90°)

Capacitor: $\mathbf{Z}_C = \frac{1}{j\omega C} = \frac{-j}{\omega C}$ (phase: -90°)

Circuit Laws (Phasors)

KVL: $\sum \mathbf{V}_k = 0$

KCL: $\sum \mathbf{I}_k = 0$

Ohm's Law: $\mathbf{V} = \mathbf{IZ}$

Series: $\mathbf{Z}_{eq} = \mathbf{Z}_1 + \mathbf{Z}_2 + \cdots$

Parallel: $\mathbf{Z}_{eq}^{-1} = \mathbf{Z}_1^{-1} + \mathbf{Z}_2^{-1} + \cdots$
Voltage divider: $\mathbf{V}_k = \mathbf{V}_s \frac{\mathbf{Z}_k}{\mathbf{Z}_1 + \mathbf{Z}_2}$

Resonance in RLC

Condition: $X_L = X_C$ or $\omega_0 L = \frac{1}{\omega_0 C}$
Resonant frequency: $\omega_0 = \frac{1}{\sqrt{LC}}$
At resonance: $\mathbf{Z} = R$ (minimum, purely resistive)

AC Power

RMS values: $V_{rms} = \frac{V_m}{\sqrt{2}}$, $I_{rms} = \frac{I_m}{\sqrt{2}}$
Average (real) power: $P = V_{rms} I_{rms} \cos\theta$ (watts)
where θ = phase angle between V and I
Reactive power: $Q = V_{rms} I_{rms} \sin\theta$ (VAR)
Apparent power: $S = V_{rms} I_{rms}$ (VA)
Power factor: $pf = \cos\theta$

Lecture 6: Diodes

Shockley Equation

$i_D = I_S(e^{v_D/nV_T} - 1)$
Parameters:

- $I_S \approx 10^{-14}$ A (saturation current)
- n = 1-2 (ideality factor)
- $V_T = kT/q \approx 25$ mV at room temp

Operating Regions

Forward bias ($v_D > 0$):

- Conducts when $v_D \geq$ 0.6-0.7 V
- Approximate as 0.7 V drop when ON

Reverse bias ($v_D < 0$):

- Tiny current $\approx -I_S$ (nA)
- Approximate as open circuit

Small-Signal Resistance

$r_d = \frac{dv_D}{di_D} = \frac{nV_T}{I_D}$
At room temp with $n = 1$:
$r_d \approx \frac{25 \text{ mV}}{I_D}$
Examples:

- $I_D = 1$ mA $\rightarrow r_d = 25\,\Omega$
- $I_D = 5$ mA $\rightarrow r_d = 5\,\Omega$

Zener Diodes

Operate in reverse breakdown at voltage V_Z
Used for voltage regulation
Maintains near-constant voltage in breakdown region

Lecture 7: Diode Applications

Half-Wave Rectifier

DC output (ideal): $V_{DC} = \frac{V_p}{\pi} \approx 0.318 V_p$
With real diode: $V_{DC} = \frac{V_p - V_D}{\pi} \approx \frac{V_p - 0.7}{\pi}$
PIV (Peak Inverse Voltage): V_p

Half-Wave with Filter

Ripple voltage: $V_r \approx \frac{V_p}{fR_L C}$
DC voltage: $V_{DC} \approx V_p - \frac{V_r}{2}$

Bridge Rectifier

DC output (ideal): $V_{DC} = \frac{2V_p}{\pi} \approx 0.637 V_p$
With real diodes: $V_{DC} = \frac{2(V_p - 2V_D)}{\pi}$ (two diode drops)
PIV: V_p (per diode)

Bridge with Filter

Ripple voltage: $V_r \approx \frac{V_p}{2fR_L C}$ (note: $2f$ due to full-wave)
Ripple is half that of half-wave rectifier!

Peak-to-RMS Conversion

$V_p = \sqrt{2} \times V_{RMS}$
$V_{RMS} = V_p/\sqrt{2}$
For 120 V AC (RMS): $V_p = 170$ V

Quick Reference

Continuity Rules

- Capacitor voltage is continuous: $v_C(0^+) = v_C(0^-)$
- Inductor current is continuous: $i_L(0^+) = i_L(0^-)$

Steady-State DC

- Capacitors: open circuit (infinite impedance)
- Inductors: short circuit (zero impedance)
- Find $x(\infty)$ by analyzing DC circuit

Frequency Behavior

Low freq ($\omega \rightarrow 0$):

- Capacitor: open ($Z_C \rightarrow \infty$)
- Inductor: short ($Z_L \rightarrow 0$)

High freq ($\omega \rightarrow \infty$):

- Capacitor: short ($Z_C \rightarrow 0$)
- Inductor: open ($Z_L \rightarrow \infty$)

Problem-Solving Steps

Time Domain (First-Order):

1. Find $x(0^+)$ using continuity
2. Find $x(\infty)$ using steady-state
3. Find τ (RC or L/R) using Thévenin
4. Apply: $x(t) = x(\infty) + [x(0^+) - x(\infty)]e^{-t/\tau}$

Frequency Domain (Phasors):

1. Convert sources to phasor form
2. Calculate impedances at given ω
3. Use circuit laws (KVL, KCL, dividers)
4. Convert back to time domain if needed

Diode Circuits:

1. Assume diode state (ON or OFF)
2. Analyze circuit with assumption
3. Check if assumption is consistent
4. If not, flip assumption and re-analyze